

AI61004 Statistical Foundations of AI/ML Quiz 1 (Set 2)

Date 12th February 2026

Time 15:00 AM -15:40 PM

Full Marks: 20

Name: _____ Roll: _____

Q1. Consider two PMFs: [0.25 0.3 0.15 0.3], and [0.1 a 0.1 b]. Calculate their entropies. For what values of 'a' and 'b' will the entropy of the second PMF be maximized? [5 marks]

Solution Sketch: This is a direct application of the definition of entropy, which is $H(p) = -\sum p_i \log(p_i)$. For the second part, it needs to be recognized that $a+b=0.8$, so there is just one free variable. The entropy must be maximized based on it, by differentiating and equating to 0.

Q2. X, Y are IID Standard Gaussian Random Variables. Calculate Moment Generating Functions of i) X^2+Y^2 , ii) XY. [5+5=10 marks]

Solution Sketch: $E(\exp(t*(X^2+Y^2))) = E(\exp(t*X^2))*E(\exp(t*Y^2))$. With a little calculation, both terms work out to $1/\sqrt{1-2t}$, so the MGF of X^2+Y^2 comes to $1/(1-2t)$.

In case of XY, $E(\exp(t*X*Y))$. Since X,Y are independent, we can first consider 'tX' as the parameter and calculate the MGF of Y, which comes to $\exp(t^2X^2/2)$. Next we calculate the MGF of X^2 considering $t^2/2$ as parameter, which we have already done, and it works to $1/\sqrt{1-t^2}$.

Q3. Prove that if $X \sim N(u, s^2)$, then $Z \sim N(0,1)$ where $Z=(X-u)/s$ [5 marks]

Solution sketch: It can be done either based on the definition of CDF, where we calculate $P(Z < t) = P(X < u + s*t)$, and then we differentiate both sides w.r.t t to get the PDF of Z, which will of course come to $1/\sqrt{2\pi} * \exp(-t^2/2)$. Alternatively, the change of variables formula can be applied on the PDF of X to get the PDF of Z, which works out to $1/\sqrt{2\pi} * \exp(-t^2/2)$. This is the PDF of $N(0,1)$. Yet another way is to calculate the MGF of Z.